

Lab: Decision Tree and Random Forest on Wisconsin Breast Dataset

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1. Presentation of the Wisconsin Breast Cancer Dataset

Overview:

- The Wisconsin Breast Cancer dataset is a widely used dataset in machine learning for binary classification tasks.
- It contains features computed from breast cancer cell nuclei images.
- The goal is to predict whether a given tumor is benign or malignant based on the features.

Dataset Details:

- **Number of Instances:** 569
- **Number of Features:** 30 (plus the target column)
- **Features:**
 - All features are real-valued, with values representing various characteristics of cell nuclei such as radius, texture, perimeter, area, smoothness, etc.
 - Some key feature categories:
 - **Radius:** Mean of distances from center to points on the perimeter
 - **Texture:** Standard deviation of gray-scale values
 - **Perimeter:** Size of the perimeter of the nuclei
 - **Area:** The area of the nuclei
 - **Smoothness:** Variation in the smoothness of the cell perimeter
 - **Compactness, concavity, and symmetry:** These also contribute to the characterization of nuclei.
- **Target Labels:**
 - **Malignant (M):** Cancerous tumor

- **Benign (B):** Non-cancerous tumor

Summary Statistics:

- **Benign Cases:** 357
- **Malignant Cases:** 212

The dataset is often used to benchmark machine learning models that can distinguish between benign and malignant tumors based on these features.

2. Lab Objectives

- Introduce students to decision trees and random forests as powerful models for classification tasks.
- Compare the performance of these models on a real-world dataset (Wisconsin breast cancer dataset).
- Analyze and interpret the results of the models.

3. Lab Structure

Part 1: Data Exploration

1. **Loading the Dataset:** Use libraries such as pandas to load the dataset.
2. **Basic Exploratory Data Analysis (EDA):**
 - Analyse basic statistics of the dataset (e.g., distribution of target labels, means, and standard deviations of features).
 - Suggest Transformations to normalize the dataset.

Part 2: Decision Trees

1. **Train a Decision Tree Model:**
 - Split the dataset into training and testing sets.
 - Train a decision tree classifier using scikit-learn.
 - Use Gini index or Entropy as criteria for splitting nodes.
2. **Evaluation:**
 - Evaluate the performance using metrics such as accuracy, precision, recall, and F1-score.

- Visualize the tree structure using tools such as graphviz or built-in scikit-learn plotting tools.

3. Hyperparameter Tuning:

- Test the effect of changing hyperparameters like tree depth, minimum samples per leaf, etc.

Part 3: Random Forest

1. Train a Random Forest Model:

- Train a random forest classifier with the same training/testing split.
- Experiment with the number of estimators (trees) in the forest.

2. Feature Importance:

- Use the feature importance attribute to find the most important features for classification.

3. Model Performance:

- Compare the performance of the random forest model with the decision tree using accuracy, precision, recall, and F1-score.
- Experiment with out-of-bag (OOB) score as an additional evaluation metric.

Part 4: Feature Importance Analysis¹

1. Feature Importance in Decision Trees:

- **Definition:** Measures how much each feature contributes to reducing impurity.
- **Procedure:**
 - Access `feature_importances_` in scikit-learn after training the model.
 - Visualize feature importance with a bar chart:

¹ https://scikit-learn.org/stable/auto_examples/inspection/plot_permutation_importance_multicollinear.html

```
import matplotlib.pyplot as plt
import numpy as np
feature_importance = model.feature_importances_
features = X.columns
plt.figure(figsize=(10, 6))
plt.barh(np.arange(len(features)), feature_importance, align='center')
plt.yticks(np.arange(len(features)), features)
plt.xlabel("Feature Importance")
plt.ylabel("Features")
plt.title("Decision Tree - Feature Importance")
plt.show()
```

Questions:

- Which features are most important for decision trees?
- Do the most important features align with breast cancer knowledge?

2. Feature Importance in Random Forests:

- **Definition:** Averages feature importance across all trees in the forest.
- **Procedure:**
 - Use the `feature_importances_` attribute of the random forest model.
 - Visualize the importance of features:

```
rf_feature_importance = random_forest_model.feature_importances_
plt.figure(figsize=(10, 6))
plt.barh(np.arange(len(features)), rf_feature_importance, align='center')
plt.yticks(np.arange(len(features)), features)
plt.xlabel("Feature Importance")
plt.ylabel("Features")
plt.title("Random Forest - Feature Importance")
plt.show()
```

Questions:

- Are the same features important in both decision trees and random forests?
- Which features are consistently ranked as highly important?

4. Questions for Analysis

Part 1: Decision Trees

1. What is the accuracy of the decision tree model on the testing set? How does it compare to the training set? Does the model overfit or underfit the data?
2. Experiment with different depths of the decision tree. What happens when you restrict the maximum depth? What are the benefits and drawbacks of doing so?
3. What criteria (Gini or Entropy) performed better? Why do you think that is the case?
4. Visualize the decision tree. Are there any features that are consistently used at higher levels of the tree (closer to the root)? Why do you think these features are important?

Part 2: Random Forest

1. How does the accuracy of the random forest model compare to the decision tree? Which model generalizes better to unseen data?
2. Investigate the impact of changing the number of trees in the random forest. Does adding more trees always improve accuracy? Why or why not?
3. Analyze the feature importance scores from the random forest. Which features are most important for predicting breast cancer? Do they align with domain knowledge or medical understanding?
4. What are the advantages of using a random forest over a single decision tree? Discuss concepts like variance reduction and robustness to overfitting.

Part 3: General Comparison

1. Compare the overall performance of the decision tree and random forest models. Which one would you recommend for this task and why?
2. Discuss the trade-offs between model interpretability (decision trees) and accuracy (random forests).
3. How could you further improve the performance of these models? Would using techniques like bagging, boosting, or more advanced models (e.g., gradient boosting) make a difference?

5. Conclusion and Discussion

- Summarize your findings from the experiments.
- You are encouraged to reflect on the importance of hyperparameter tuning, model evaluation, and the concept of bias-variance tradeoff.
- Optionally, you can apply cross-validation to further validate the stability of your models.